

PHYO THANT KYAW

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github.com/pt22-mfu | pt22-mfu.github.io/pt_portfolio | [LinkedIn](#)

Nationality: Myanmar | Visa: Non-Immigrant ED Plus | Authorized: 16-week Co-op, No Sponsorship Required

Eligible: 1-Year Post-Grad Extension + BOI IT Specialist Tier | Languages: English (Professional), Thai (Basic), Burmese (Native)

SUMMARY

Data engineering specialist with expertise in designing automated ETL pipelines, complex feature engineering, and Python-based data infrastructure on real-world environmental datasets — with hands-on experience debugging and shipping production deployments end-to-end.

Seeking a 16-week Co-op (January 2027) to architect scalable data infrastructure and optimize end-to-end data pipelines at a Tier-1 Thai tech organization.

EDUCATION

Mae Fah Luang University (MFU) — B.Eng., Computer Engineering GPAX: 3.84 | Expected 2027

Mandatory 16-week Cooperative Education: January 2027

Mandalay Technological University (MTU) — B.Eng., Electronics Engineering GPA: 3.04 | Dec 2016 – Sep 2019 (Year 3)

TECHNICAL PROJECTS

AI Vision QC Inspector [Live Demo](#) | [GitHub](#)

Aug 2026

- Built an automated inference pipeline (image ingestion → preprocessing → model prediction → structured output) achieving 89.4% validation accuracy on 6,633 training images.
- Designed the data flow connecting model output to a GenAI reporting layer (Gemini 2.5 Flash), turning raw predictions into structured, actionable reports.
- Diagnosed and resolved production infrastructure issues end-to-end: Python/runtime version incompatibility, dependency resolution conflicts, and deprecated SDK migration on Streamlit Cloud.

MFU PM2.5 GeoAI Warning System — Senior Project [Live Demo](#) | [GitHub](#)

Jan – Jul 2026

- Architected a 4-model comparative forecasting pipeline (LightGBM, XGBoost, SVR, MLR) using a strict chronological train/validation/test split, with LightGBM emerging as champion at 85.90% R^2 (MAE: 3.21) against a 5-year localized Chiang Rai dataset.
- Engineered distance-decay Fire Pressure features across 25km/50km/100km radiuses from NASA FIRMS (VIIRS) satellite hotspot data, integrated with OpenWeatherMap and Air4Thai meteorological APIs.
- Implemented data validation checkpoints and a graceful fallback architecture across the ingestion pipeline, ensuring stable data delivery during API downtime.

Chiang Mai Tri-Node Zone: Geo-Predictive AI & Mitigation Platform [Live Demo](#) | [GitHub](#)

Mar – May 2026

- Engineered Z-score spatial analytics and wind-direction correlation pipelines across 3 Chiang Mai monitoring nodes to detect micro-level air quality shifts in real time.
- Transformed raw multi-node NASA FIRMS (Fire Radiative Power) and OpenWeatherMap records using Pandas — standardizing coordinate systems, resolving missing values, and structuring outputs for LLM prompt injection.
- Built robust Python data delivery pipelines feeding an interactive geospatial threat-visualization dashboard with zero manual refresh.

EXPERIENCE

Work Assistant — Web & Data Projects — MFU Learning Innovation Institute (MLii) *May 2026 – Jul 2026*

- Engineered a secure, Thai PDPA-compliant PostgreSQL database architecture via Supabase to ingest, encrypt, and store clinical assessment data.
- Architected a backend data flow integrating a Zero-Budget Hybrid AI Engine (Rule-based + Gemini LLM), optimizing API response times with Upstash Redis caching.
- Authored the complete Software Requirements Specification (SRS), structuring RESTful API endpoints, robust data validation pipelines (Zod), and Role-Based Access Control (RBAC).

TECHNICAL SKILLS

Data Engineering: ETL Pipeline Design, Data Preprocessing, Feature Engineering, API Integration (REST), Data Validation, Apache Spark, Hadoop

Programming & DB: Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL, TypeScript

Cloud & Infrastructure: Google Cloud Platform (GCP), Docker, Git, GitHub, GitHub Actions (CI/CD)

Analytics & AI: LightGBM, XGBoost, TensorFlow, Streamlit, LangChain, Google Gemini API

Certifications: IBM Data Engineering Professional | Google Data Analytics | Google Cloud: Data Lakes & Warehouses | UC Davis: SQL for Data Science | Google AI Essentials, Prompting Essentials